Colloids and sedimentation

Complex disordered systems: Colloids

A particle suspended in a liquid can be defined as *colloidal* if the kinetic energy of its thermally-driven Brownian motion is large enough to prevent it from settling completely to the bottom of the container under the influence of gravity.

A second definition of a colloid is a particle whose gravitational Peclet number Pe is significantly smaller than 1. (The gravitational Peclet number is the ratio of the time taken by a particle to diffuse a distance equal to its radius to the time taken by it to sediment the same distance under gravity.)

Show that these two definitions of a colloid are essentially equivalent.

Solution

To show the equivalence between these two definitions, we need to establish the relationship between thermal energy dominating over gravitational settling and the gravitational Peclet number being much less than 1.
For the first definition, a spherical particle of radius a and density ρ_p in a fluid of density ρ_f, the gravitational force is:

$$F_g = \frac{4\pi a^3}{3}(\rho_p - \rho_f)g$$

The gravitational potential energy difference over a distance equal to the particle radius is:

$$U_g = F_g \cdot a = \frac{4\pi a^4}{3}(\rho_p - \rho_f)g$$

For thermal motion to dominate, we require:

$$k_B T \gg U_g = \frac{4\pi a^4}{3}(\rho_p - \rho_f)g$$

For the second definition, the gravitational Peclet number is defined as:

$$\mathrm{Pe} = \frac{t_{\text{diffusion}}}{t_{\text{sedimentation}}}$$

For diffusion over distance a:

$$t_{\text{diffusion}} = \frac{a^2}{D}$$

where D is the diffusion coefficient given by the Stokes-Einstein relation: $D = \frac{k_B T}{6\pi\eta a}$.
For sedimentation over distance a under Stokes drag:

$$t_{\text{sedimentation}} = \frac{a}{v_s}$$

where the sedimentation velocity is: $v_s = \frac{F_g}{6\pi\eta a} = \frac{2a^2(\rho_p - \rho_f)g}{9\eta}$.

Therefore:

$$\text{Pe} = \frac{a^2/D}{a/v_s} = \frac{av_s}{D} = \frac{a \cdot \frac{2a^2(\rho_p - \rho_f)g}{9\eta}}{\frac{k_B T}{6\pi\eta a}}$$

Simplifying:

$$\text{Pe} = \frac{2a^3(\rho_p - \rho_f)g \cdot 6\pi\eta a}{9\eta \cdot k_B T} = \frac{4\pi a^4(\rho_p - \rho_f)g}{3k_B T}$$

From the first definition, colloidal behavior requires:

$$k_B T \gg \frac{4\pi a^4}{3}(\rho_p - \rho_f)g$$

This is equivalent to:

$$\frac{4\pi a^4(\rho_p - \rho_f)g}{3k_B T} \ll 1$$

But this is exactly the condition $\text{Pe} \ll 1$ from the second definition.
Therefore, the two definitions are mathematically equivalent: thermal energy dominance over gravitational settling occurs precisely when the gravitational Peclet number is much smaller than unity. Both conditions ensure that Brownian motion prevents complete sedimentation, which is the hallmark of colloidal behavior.